Jeanne Astier

CREST, Institut Polytechnique de Paris, 5 avenue Henry Le Chatelier, 91120 Palaiseau, France
jeanne.astier@ensae.fr • jeanneastier.github.io • Mobile: +33 6 04 41 41 95 • Citizenship: French

Placement Director: Professor Xavier D'Haultfoeuille – xavier.dhaultfoeuille@ensae.fr

Graduate Director: Professor Alessandro Riboni – alessandro.riboni@gmail.com

Research fields

Primary: International Trade
Secondary: Environmental Economics

Education

Fall 2026	**Globalization Ph.D. Fellow, Dartmouth College** part of the Dartmouth globalization group for a semester
Spring 2026	**Visiting Ph.D. student, ARE Department, UC Berkeley** hosted by Prof. Thibault Fally
2023 – 2027	**Ph.D., Economics, CREST, Institut Polytechnique de Paris** advised by Prof. Geoffrey Barrows and Prof. Florian Grosset-Touba
2022 – 2023	**M.S., Economics, Institut Polytechnique de Paris**
2018 – 2023	**M.M., Management, HEC Paris**
2018 – 2023	**Engineering Degree, Statistics, ENSAE Paris**
2019 – 2020	**B.S., Modern Literature, Université Paris 3**

References

Professor Geoffrey Barrows (main advisor)
CREST, École Polytechnique – CNRS
geoffrey-masters.barrows@polytechnique.edu

Professor Treb Allen
Department of Economics, Dartmouth College
treb@dartmouth.edu

Professor Cécile Gaubert
Department of Economics, University of California, Berkeley
cecile.gaubert@berkeley.edu

Professor Thibault Fally
Department of Agricultural and Resource Economics, University of California, Berkeley
fally@berkeley.edu

Working papers

Climate-Induced Congestion in Ports: General Equilibrium Consequences on Transportation and Trade

Job Market Paper

Abstract: How does climate change affect the cost of international shipping? I show that the maritime shipping network amplifies local port disruptions into large global costs. Rising sea levels threaten port operations, and heterogeneous climate dynamics make the threat uneven across ports. Because container shipping operates as a global network, the damage propagates well beyond the ports directly exposed. I combine high-frequency satellite shipping data with sea weather observations to estimate how high water levels disrupt port operations. I embed these estimates in a general equilibrium model of international trade with a shipping network. At ports, higher water levels reduce capacity and trigger congestion, so costs rise with traffic. On shipping links, by contrast, economies of scale from cargo pooling make costs fall with traffic and rationalize the network's sparsity – allowing the network structure to respond endogenously to changes in climate conditions. Sea level rise projections imply that ports will face water level increases of up to 1.3 meters by the end of the century. I find that such increases raise trade costs by 4.6% to 7.2% on average across country pairs and lower global welfare by 1.4% to 2.0%, depending on the climate scenario. These results add a new component to the climate damage function: the sensitivity of trade costs to climate change. In a sparse and concentrated shipping network, the direct effect of sea level rise is dominated by reallocation forces: a country's welfare loss depends more on its position in the trade network than on the exposure of its own ports.

Quantifying Climate Damages When Regions Trade: A Structural Gravity Approach

with Geoffrey Barrows (CREST - CNRS), Raphael Calel (Georgetown University), Hélène Ollivier (PSE - CNRS)

Abstract: This paper presents a method for estimating treatment effects of local climate shocks when regions trade. Because trade creates spillovers, comparing outcomes across regions with different shock exposure leads to bias. We model these spillovers using standard assumptions from international trade theory and develop a model-consistent strategy for estimating key parameters and deriving counterfactuals. Unlike existing quantitative trade approaches, however, we leverage similar identifying variation as reduced-form estimators, eliminating the need for auxiliary data on productivity, prices, or quantities. Using our strategy, we revisit the literature on climate change and gross output, finding substantially larger climate damage projections.

Work-in-progress

Monadic Cost Shifters in Gravity Models: A New Estimator with an Application to Environmental Regulation

with Geoffrey Barrows (CREST - CNRS) and Hélène Ollivier (PSE - CNRS)

Tonnage Taxation in the Global Shipping Industry

with Paul-Emmanuel Chouc (CREST, Institut Polytechnique de Paris) and Sandra Poncet (Paris School of Economics)

Maritime Shipping and Ocean Biodiversity

with Chloé Antoine (CREST, Institut Polytechnique de Paris)

Policy reports

Housing Energy Performance and Energy Consumption: Insights from Banking Data (in French)
with Gabrielle Fack, Julien Fournel, Flavie Maisonneuve and Ariane Salem – Focus #103, Conseil d'Analyse Economique, 2024

Measuring the heterogeneous effects of inflation on households (in French)
with Xavier Jaravel and Madeleine Péron – Focus #99, Conseil d'Analyse Economique, 2023

Talks

2026	IP Paris Seas and Ocean Forum (poster), UC Berkeley Trade Lunch, UC Davis Nature Policy Lab, UC Berkeley Climate Change Economics Lunch (egg timer), UC Berkeley Environmental, Energy, and Resource Economics seminar, Midwest International Trade & Theory Conference (Ohio State), Rocky Mountain Empirical Trade Conference, Toulouse 15th Energy and Climate Economics Conference, RIEF network conference, Second Paris-Saclay Conference on Trade and the Environment, Dartmouth Trade Internal Seminar
2025	CREST PhD seminar, Columbia IPWSD, PSE Doctorissimes, LSE Workshop for Early Career Women In Economic Geography and Spatial Economics, LSE Environment Camp, Warwick PhD Conference, EAERE Annual Conference (Bergen), Vienna Workshop in International Economic Networks, JIE Summer School (poster), LSE Environment Week, Kiel trade talk (online), PSE Globalization Shipping and Trade workshop, Crest-Insee workshop, CEPR Paris Symposium (poster)
2024	FAERE PhD workshop (Annecy), EAERE Early Career workshop (L'Aquila), CREST PhD workshop

Grants and awards

2026	Globalization PhD Fellowship Grant, Dartmouth College
2026 – 2027	4th year PhD grant, CREST
2026	Visiting grant, ENSAE Paris
2025	Travel grant, ADRES
2023 – 2026	Teaching assistant scholarship, ENSAE Paris
2023 – 2026	Full Ph.D. scholarship, ENSAE Paris

Teaching

fall 2025	**Microeconomics 1** tutorials, ENSAE graduate students, Prof. Philippe Choné
fall 2025	**Microeconomics 2AD/3A-CI** tutorials, ENSAE graduate students, Prof. Lionel Wilner and Rémi Avignon
spring 2025	**Introduction to Environmental Economics** tutorials, ENSAE graduate students, Prof. Florian Grosset-Touba
spring 2024, 2025	**Introduction to Microeconomics** tutorials, ENSAE undergraduate students, Prof. Laurent Linnemer
fall 2023, 2024	**Univariate Descriptive Statistics** tutorials, ENSAE undergraduate students, Prof. Thierry Kamionka

Professional activities

Seminar and Conference Organization

2025	Co-organizer of PSE Globalization, Shipping, and Trade Workshop
2024, 2025	Co-organizer of CREST PhD Seminar

Referee: *Economic Journal*

Selected work experience

Jun. – Sept. 2023	**Research assistant, French Council of Economic Analysis (CAE)** for Camille Landais, Madeleine Péron, Ariane Salem
Jan. – Jul. 2022	**Research assistant, French Treasury** in the Public Policies Service – Redistribution bureau
Jul. – Dec. 2021	**Research assistant, European Central Bank** in the Directorate General Research – Monetary Policy Research Division

Skills

Languages: French (native), English (fluent), Spanish (intermediate)

Programming: Stata, Python (advanced) ; Matlab, R (intermediate)

Last updated: October 21, 2026